higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 3

7 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

- Make sure that electrical wiring and equipment are correctly maintained and correctly installed.
- Make sure that there are not any loose, exposed connections which may cause a dangerous situation.
- Make sure that all cables are correctly insulated.
- Switch off and unplug all electrical tools and appliances when they are not in use.
- Never overload plugs.
- Make sure that electrical appliances are always in a secure and safe place.
- Avoid placing light bulbs or heaters near flammable materials. (Any 5 × 1) **[5]**

QUESTION 2

2.1.1 True
2.1.2 True
2.1.3 True
2.1.4 True
2.1.5 False

(5 × 1) (5)

2.2
- Shoulders can be injured
- Hands can be injured
- Lower back can be injured
- Knees can be injured (4)

2.3
- Stop
- Emergency stop
- Lower the load
- Lift the load
- Move slowly
- Move to the left
- Move to the right
- Extend the boom
- Retract the boom
- Finished with the crane (Any 7 × 1) (7)

2.4
- Bowline
- Sheet bend
- Round turn and two half hitches
- Figure-eight knot (4)

[20]

QUESTION 3

3.1 Force acting over a distance✓ which causes an object to move.✓ (2)

3.2
- Thermal energy is the energy contained by an object✓ in the form of heat.✓
- Kinetic energy is the energy contained by an object✓ to keep it in motion.✓ (2 × 2) (4)

3.3
3.3.1 H
3.3.2 B
3.3.3 C
3.3.4 D
3.3.5 E
3.3.6 F
3.3.7 G
3.3.8 A
(8 × 1) (8)

3.4
3.4.1 Carbon brushes
3.4.2 Pivot
3.4.3 Rotation produced
3.4.4 North/N
3.4.5 South/S
3.4.6 Split-ring commutator
(6 × 1) (6)
[20]

QUESTION 4: TRANSMISSIONS

4.1 This is the system that channels power from the motor or the engine✓ to the moving parts of the machine✓ to allow the machine to work.✓ (3)

4.2
- Cover plate
- Pressure plate
- Release plate
- Diaphragm spring
- Clutch plate (5)

4.3 The brake pads are cheaper to replace. (1)

4.4 Used in:
- Train locomotives
- Ships (Any ONE) (1)
[10]

QUESTION 5: MAINTENANCE OF VEHICLES

5.1
- Make sure the engine of the vehicle is quite warm.
- Remove any dirt from around the sump plug.
- Use a spanner to remove the sump plug and drain the old oil into a catch pan.
- When all the oil has been drained, replace the sump plug.
- Use a spanner to remove the old oil filter, drain any remaining oil into the catch pan and then insert a new oil filter using a spanner to tighten it.
- Remove the filter cap at the top of the engine and add the correct quantity of oil.
- Replace the filler cap. (7)

5.2
5.2.1 300 mm
5.2.2 1,8 m
5.2.3 Green light
5.2.4 Yellow light
5.2.5 Red light
(5 × 1) (5)

5.3
5.3.1 Check daily

5.3.2 Check daily

5.3.3 Check daily

5.3.4 Check daily

5.3.5 Check weekly

5.3.6
- After every 10 000 km–15 000 km
- According to the manufacturer's instructions (Any ONE)

(6 × 1) (6)

5.4
- Jack
- Wheel spanner (2)

[20]

QUESTION 6: SMALL PLANT AND POWER TOOL OPERATION

6.1
- Generators are used to provide electric power in places where electricity from a main supply cannot be provided.
- Generators are used where electric power is temporarily needed. (2)

6.2
- It will strain the mixer.
- The motor of the mixer can burn out.
- The mixture will not be properly mixed/mixture will be unworkable.
- The mixture will not be strong enough for the construction job.
- The concrete mixer can topple over and injure persons standing nearby.

(Any 4 × 1) (4)

6.3
- Make sure that the tool is correctly wired.
- Before switching it on, make sure that its own switch is off.
- Make sure that all the accessories are properly adjusted before switching it on.
- Do not operate in wet conditions.
- Plug it out if not in use.
- Ensure that there is enough light when operating the tool.
- Read and understand the manufacturer's manual for that tool before operating it.
- Remove all the guards and store them in a safe place after use.(Any 6 × 1) (6)

6.4
- Motor
- Electric cord
- Switch
- Grinding disc
- Safety guard
- Handle (Any 3 × 1) (3)

[15]

QUESTION 7: OXYACETYLENE

7.1 It is an appliance✓ that is used to control pressure✓ (2)

7.2
- Gas cylinder
- Gas regulator
- Torch
- Nozzles
- Flashback arrestors (Any 4 × 1) (4)

7.3
- Cylinders must be stored in a dry and well-ventilated area.
- Cylinders must be stored in a clean, dirt-free area away from combustible materials.
- The gas storage area must be well-labelled with correct signs.
- Empty containers should be well-marked and chained separately from full containers. (4)

[10]

TOTAL: 100

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